

Disorders of The Respiratory System



BY

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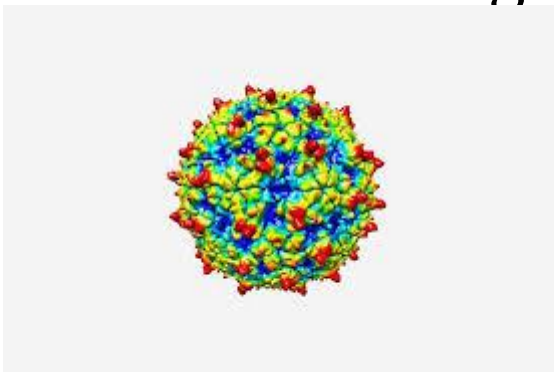
Common Cold

• Common cold is the most frequently occurring illness in childhood. It is more common in autumn and winter.



ETIOLOGY:

• More than 200 viruses can cause common cold e.g. rhinoviruses.



Predisposing factors:

• Overcrowding, passive smoking and attending daycare sittings.



Common Cold



CLINICAL MANIFESTATIONS:

In infants

- Fever, irritability.
- Feeding and sleeping difficulties duo to nasal obstruction.
- Cough which may lead to vomiting and mild diarrhea.

In older children

- It starts with nasal irritation then nasal secretion, sneezing and sore throat.
- Low grade fever, myalgia, headache, decreased appetite and cough.

Common Cold

Complications:

- ✓ Secondary bacterial infection.
- ✓ Asthma may be triggered by common cold.

TREATMENT:

- ➢ No specific treatment.
- ➢ Acetaminophen or ibuprofen for fever.
- ➢ Saline nasal drops.
- ➢ No antibiotic is recommended.
- ➢ High dose of vitamin C.



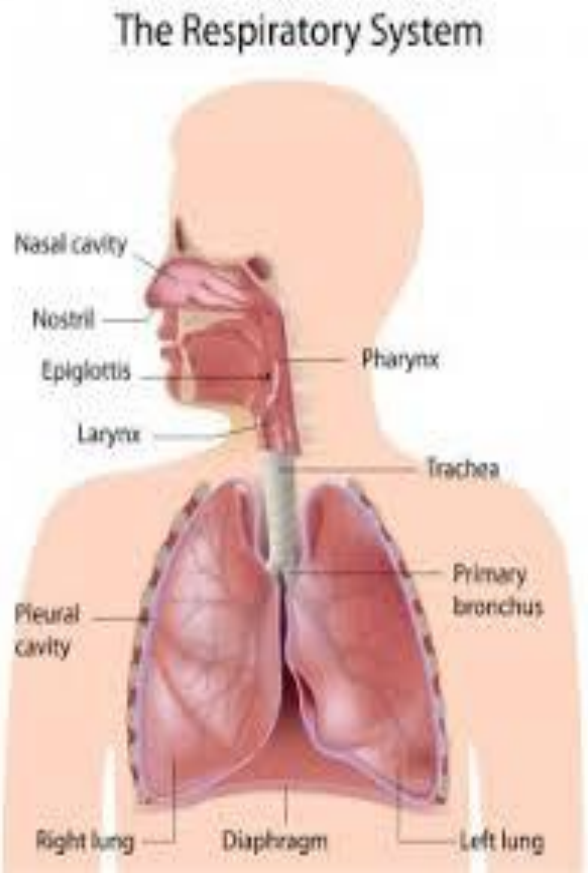


Pneumonias

Definition: Inflammation of the parenchyma of the lung.

- It is the main cause of respiratory distress and hospital admissions.







Pneumonias

Classification:

I-Anatomical

1- Lobar pneumonia: inflammation Is localized one or more lobes of the lung. It is almost always due to bacteria. Usually unilateral.

2- Lobular pneumonia (bronchopneumonia): leads to patchy inflammation in the lungs. It is usually bilateral. It is bacterial or viral in origin.

3- Interstitial pneumonia: it is an inflammation of the interstitium of the lungs, usually bilateral, and mainly due to viral infection. Usually bilateral

Pneumonias

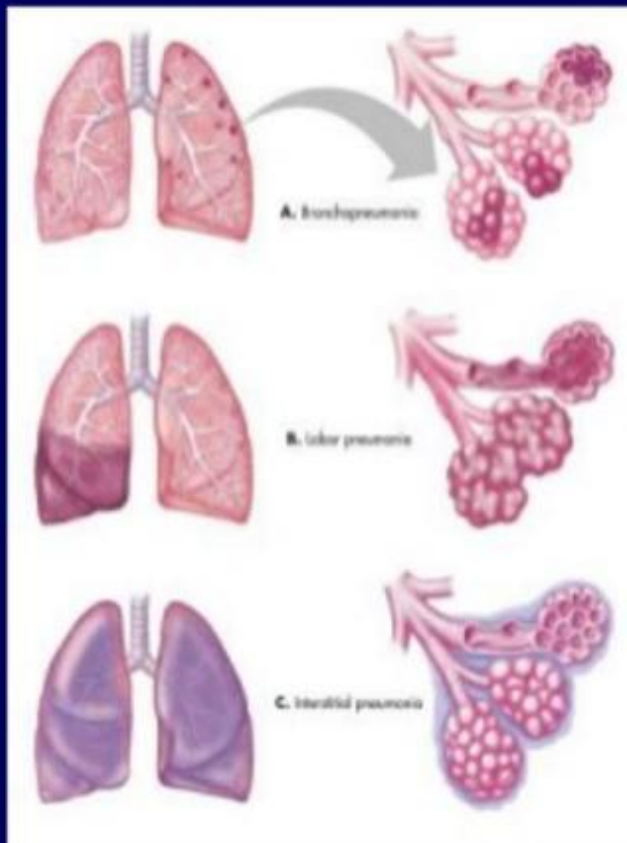
Classification:

ANATOMICAL CLASSIFICATION

1. Bronchopneumonia affects the lungs in patches around bronchi

1. Lobar pneumonia is an infection that only involves a single lobe, or section, of a lung.

1. Interstitial pneumonia involves the areas in between the alveoli



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Pneumonias

II- Etiological classification:

1- Infectious: 90% of cases.

a- Bacterial: e.g. pneumococcus, hemophilus influenza, staphylococcus, and TB.

b- Viral: e.g. respiratory syncytial virus, influenza virus, and parainfluenza virus.

c- Protozoal

d- Mycotic (fungal)

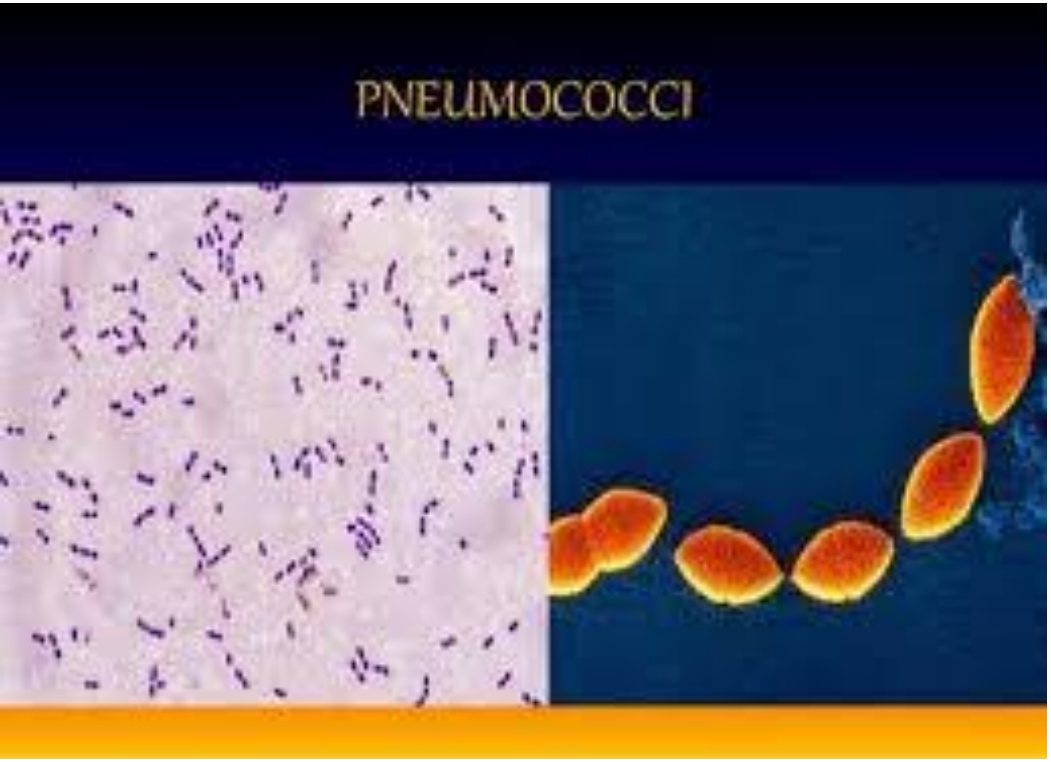
2- Non Infectious: includes:

a- Aspiration pneumonia: e.g. aspiration of amniotic fluid and gastric contents.

b- Chemical pneumonia.

Pneumococcal pneumonia

- Caused by streptococcus pneumoniae, it represents 90% of childhood bacterial pneumonia, and occurs mainly in winter and early spring.





Pneumococcal pneumonia

Clinical manifestations

- After a brief mild upper respiratory infection, there is often onset of a shacking chill followed by fever as high as 40 C. this is accompanied by rapid respiration and manifestations of respiratory distress, dry cough and anxiety.
- The child prefers to lie on the affected side to minimize chest pain and improve ventilation.

Pneumococcal pneumonia



Diagnosis:

1- Chest X-ray.

2- CBC: leukocytosis.

3- Isolation of the organism from secretions obtained by deep cough or tracheal aspiration.

4- Blood culture: bacteremia is found in 30% of patients.

Pneumococcal pneumonia

Complications:

I- Pulmonary:

1- Lung Collapse

2- Lung abscess

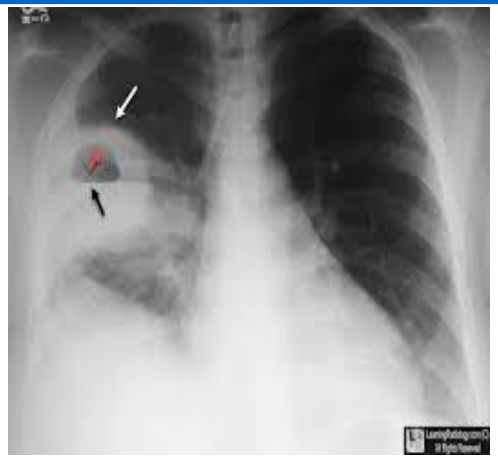
3- Pleural effusion (fluid in the pleural space)

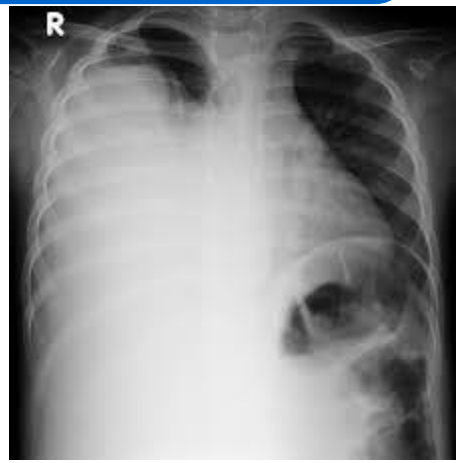
4- Empyema (pus in the pleural space)

5- Pneumothorax (air in the pleural space)

6- Respiratory failure











II- Extra Pulmonary complications:

1- Septicemia, meningitis, arthritis, osteomyelitis, and peritonitis

2- Heart failure.

Pneumococcal pneumonia

Treatment:

Hospitalization is indicated in severe respiratory distress.



1- Antibiotics

2- Oxygen therapy

3- Symptomatic treatment e.g. antipyretics

4- Good nutrition



5- Oral zinc

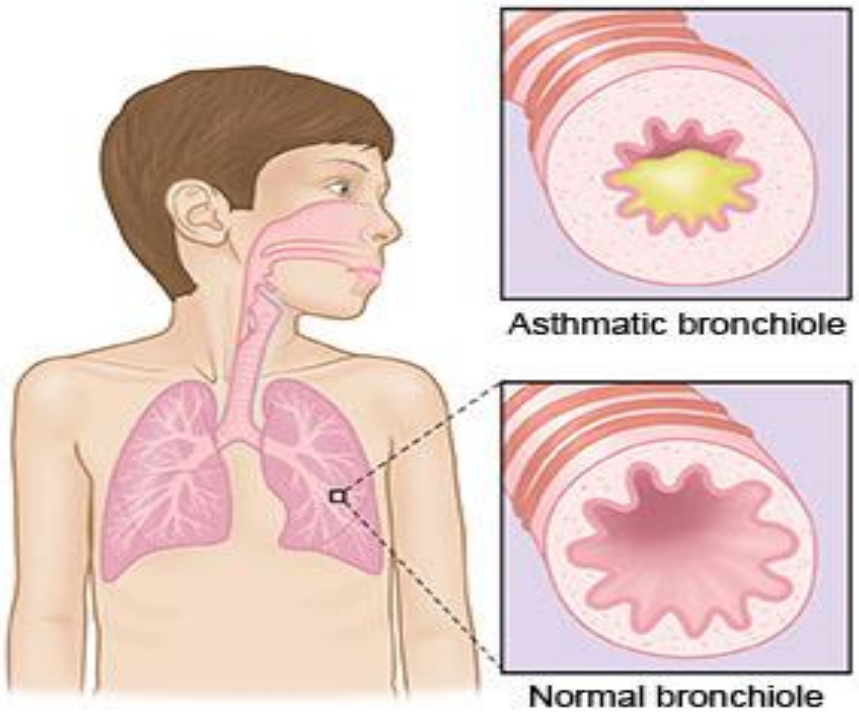
6- Treatment of complications

Bronchial Asthma

Definition: chronic inflammatory disease with diffuse and reversible bronchoconstriction in response to a variety of stimuli.

Etiology: asthma is usually a complex disorder involving immunologic, infectious, and psychologic factors.





Bronchial Asthma

❖ **Asthma triggers:**

1- Viral infection of the respiratory tract.

2- Animal dander

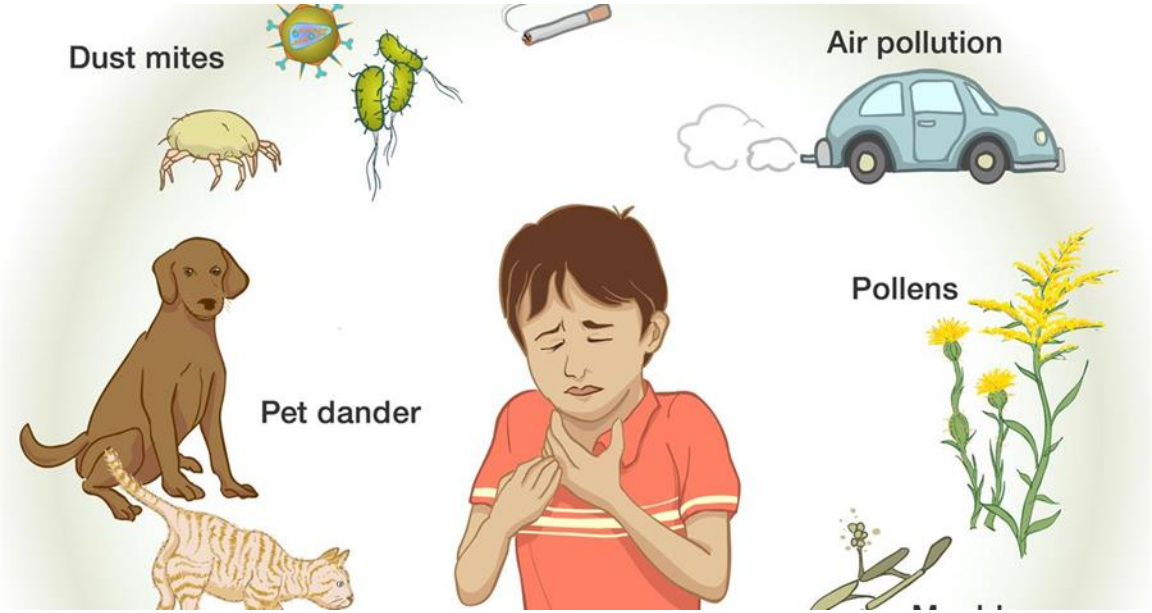
3- Dust mites

4- Pollens

5- Tobacco smoke

6- Air pollutants.

7- Strong odors e.g. perfumes



Bronchial Asthma

Clinical manifestations:

1- Intermittent dry cough and/ or wheeze are the most common Symptoms.

2- Shortness of breath.

3- Symptoms are worse at night

4- Other allergic conditions e.g. eczema





Bronchial Asthma

Diagnosis:

1- Clinical manifestations: recurrent episodes of cough and/ or Wheeze triggered by exercise, viral infection or inhaled allergens

2- + ve family history of asthma

3- Improvement after use of bronchodilators







Bronchial Asthma

4- **Investigations:**

- Pulmonary function test
- Chest X-ray
- Eosinophilia
- Allergic skin tests may identify asthma triggers







Bronchial Asthma

Treatment:

1- Quick relief medications (relievers): are taken on occurrence of the attack usually by inhalation e.g. salbutamol

2- Long-term control medications (controllers): are taken by inhalation or orally e.g. steroids and theophylline.






Thank You!